INFO 201 Final Project BC-6

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Introduction

Suicide is one of the leading causes of death globally, with over 720,000 lives lost each year according to the World Health Organization (WHO). This project seeks to uncover the underlying factors contributing to suicide rates by analyzing trends, demographic disparities, and socio-economic influences. Using data from WHO Suicide Statistics and another dataset on global suicide, economic conditions, and substance use disorders, we aim to answer the following questions:

- How do suicide trends differ by sex over time globally?
- How does economic hardship influence suicide rates across different countries? What are the patterns in male and female suicide rates in high-suicide-rate countries?
- How do suicide rates from the two datasets compare? Is one of them underreported?

Our goal is to provide insights that can help inform prevention strategies and policy interventions by visualizing and analyzing suicide trends globally.

Ethical Considerations

- Underreport Bias: Suicide data may be underreported due to cultural stigma or inconsistent record-keeping in certain countries.
- Data Gaps: Some countries or demographics lack comprehensive data, which can skew trends or introduce bias.
- Cultural Differences: Cause of suicide may vary significantly across regions due to unique cultural, social, and economic contexts, which general analyses may fail to capture.

Data Source

1. WHO Suicide Dataset: https://www.kaggle.com/datasets/dhanushkatharanga/who-suicide-data Collected by the World Health Organization,contains information on suicide counts, populations, and demographics across various countries and years.

2. Global Suicide, Mental Health, Substance Use Dataset: https://www.kaggle.com/datasets/thedevastator/global-suicide-mental-health-substance-use-disor Provides standardized suicide rates per 100,000 population for different genders and age groups.

Both datasets were merged using shared keys (country, year, sex) to create a comprehensive dataset for our analysis.

Loading Libraries

```r
library(dplyr)
library(stringr)
library(ggplot2)
library(tidyr)
library(plotly)
```

Loading Data Sets

```r
suicide_df <- read.csv("who_suicide_statistics.csv")
crude_suicide_df <- read.csv("crude suicide rates.csv")
```

Data Cleaning

```r
colnames(crude_suicide_df) <- c(
    "country", "year",
    "Both Sexes None suicide (30-70)",
    "Male None suicide (30-70)",
    "Female None suicide (30-70)",
    "Both Sexes Crude suicide",
    "Male Crude suicide",
    "Female Crude suicide"
    )
glimpse(crude_suicide_df)
```

```
Rows: 916
Columns: 8
$ country                           <chr> "Country", "Afghanistan", "Afghanist…
$ year                              <chr> "Year", "2016", "2015", "2010", "200…
$ `Both Sexes None suicide (30-70)` <chr> "Both sexes", "29.8", "29.8", "31.7"…
$ `Male None suicide (30-70)`       <chr> "Male", "31.8", "31.9", "34.1", "36.…
$ `Female None suicide (30-70)`     <chr> "Female", "27.7", "27.8", "29.4", "3…
$ `Both Sexes Crude suicide`        <chr> "Both sexes", "4.7", "4.8", "5.1", "…
$ `Male Crude suicide`              <chr> "Male", "7.6", "7.8", "8.6", "10.8",…
$ `Female Crude suicide`            <chr> "Female", "1.5", "1.5", "1.4", "1.5"…
```

Column Information/Codebook

WHO Suicide Dataset:

1. country: The name of the country where the data was collected.
2. year: The calendar year of the data collection.
3. age: Age group of the population (e.g., “25-34 years”).
4. sex: Gender of the population (Male, Female, Both sexes).
5. suicides_no: The absolute number of suicides recorded for the specified group.
6. population: The total population count for the specified group (age, sex).
7. suicide_rate (calculated): The suicide rate per 100,000 population, derived using suicides_no / population * 100,000.

Crude Suicide Rates Dataset:

1. country: The name of the country where the data was collected.
2. year: The calendar year of the data collection.
3. Both Sexes None suicide (30-70): Probability of dying from causes other than suicide between ages 30-70, aggregated for both sexes.
4. Male None suicide (30-70): Probability of dying from causes other than suicide between ages 30-70 for males.
5. Female None suicide (30-70): Probability of dying from causes other than suicide between ages 30-70 for females.
6. Both Sexes Crude suicide: Crude suicide rate per 100,000 population for both sexes.
7. Male Crude suicide: Crude suicide rate per 100,000 population for males.
8. Female Crude suicide: Crude suicide rate per 100,000 population for females.
9. Measure (after transformation): Indicates the type of metric (e.g., “Crude suicide”).

We will join the datasets based on the common columns of country and year, as they are serve as the primary keys in both datasets.

The probability of dying between the ages of 30 and 70 from any of cardiovascular disease, cancer, diabetes, or chronic respiratory disease. (Percentage)

Crude Suicide Rates (per 100 000 Population)

```
crude_suicide_df <- crude_suicide_df |>
    slice(-1)
  crude_suicide_df$year <- as.integer(crude_suicide_df$year)
  crude_suicide_df <- crude_suicide_df|>
    pivot_longer(
      !c(country, year),
      names_to = c("sex", "Measure"),
      names_pattern = "(Both Sexes|Male|Female) (.*)"
      )
```

Filtering Age Group to apporximately match with the age range in WHO dataest

```
suicide_aged_df <- suicide_df[suicide_df$age %in% c("25-34 years", "35-54 years", "55-
```

Calculating suicide rate per 100 000 population from WHO dataset

```
suicide_sex_df <- suicide_aged_df |>
    group_by(country, year, sex) |>
    summarize(
      population = sum(population, na.rm = TRUE),
      suicides_no = sum(suicides_no, na.rm = TRUE)
    ) |>
    ungroup() |>
    mutate(suicide_rate = (suicides_no/population)*100000)
```

```
`summarise()` has grouped output by 'country', 'year'. You can override using
the `.groups` argument.
```

Calculating the suicide rate per 100,000 population allows for a standardized comparison of suicide prevalence across countries and years, irrespective of population size. This ensures the data is comparable and highlights trends or disparities in suicide rates more effectively.

Group by country and year, and calculate "Both sexes" population and suicide numbers

```
suicide_both_sexes_df <- suicide_sex_df |>
   group_by(country, year) |>
   summarise(
     sex = "Both sexes",
     population = sum(population, na.rm = TRUE),
     suicides_no = sum(suicides_no, na.rm = TRUE),
     .groups = "drop"
   ) |>
   mutate(suicide_rate = (suicides_no/population)*100000)
```

Grouping by country and year and calculating "Both sexes" population and suicide numbers provides a comprehensive overview of overall suicide rates without the gender-specific breakdown. This aggregation simplifies cross-country or temporal comparisons while preserving critical population-level insights.

Combine the "Both sexes" data with the original dataset

```
suicide_rate_final_df <- bind_rows(suicide_sex_df, suicide_both_sexes_df) |>
    arrange(country, year, sex)

  suicide_rate_final_df <-
    suicide_rate_final_df |>
    mutate(sex = ifelse(sex == "female", "Female", sex))

  suicide_rate_final_df <-
    suicide_rate_final_df |>
    mutate(sex = ifelse(sex == "male", "Male", sex))
```

Combining the "Both sexes" data with the original dataset ensures that both aggregated and gender-specific data are available in a single dataset. This facilitates flexible analysis, allowing comparisons between overall trends and gender-specific patterns.

Data Joining

```
crude_suicide_only <- crude_suicide_df |>
   filter(Measure == "Crude suicide")

 inner_join <- inner_join(suicide_rate_final_df, crude_suicide_only, by = c("country"
 left_join <- left_join(suicide_rate_final_df, crude_suicide_only, by = c("country",
```

We joined the datasets based on the columns country, year, and sex because these serve as the primary keys in both datasets. This ensures accurate alignment of data regarding suicide rates across different sexes, years, and countries.

We performed two types of joins: 1. Inner Join: To keep only the rows where there are matching records in both datasets. 2. Left Join: To retain all rows from the suicide_rate_final_df, regardless of whether there is a match in the crude_suicide_only dataset.

Graphing

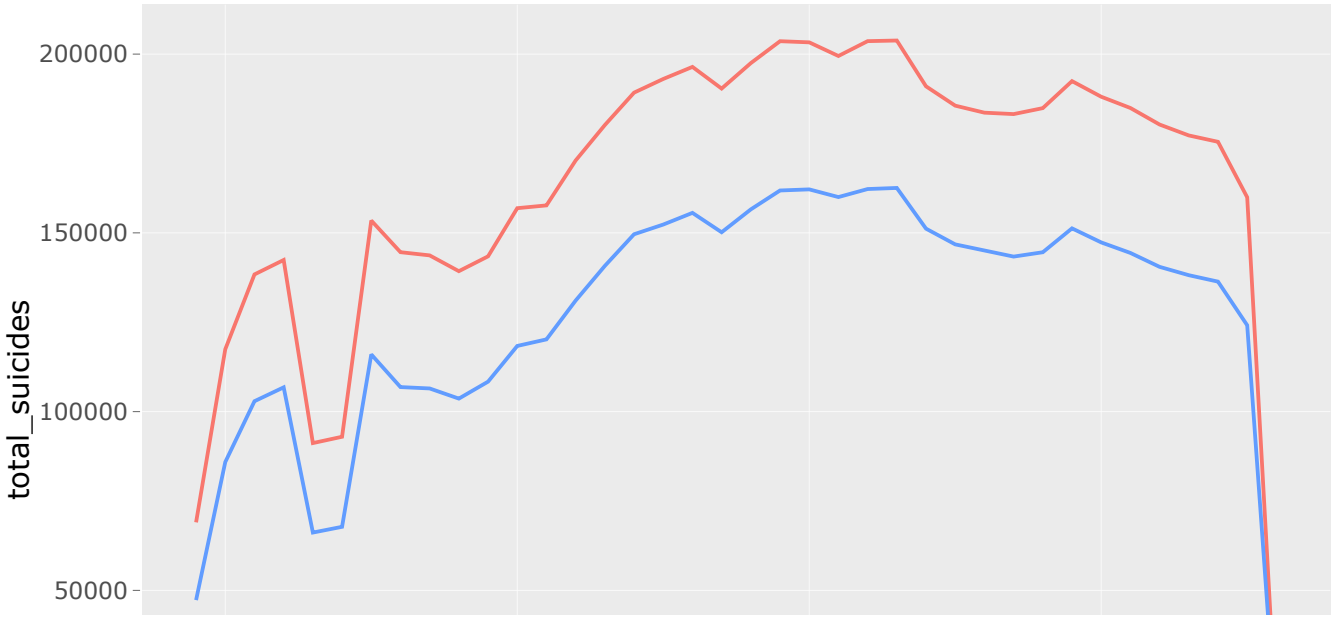
Graph 1

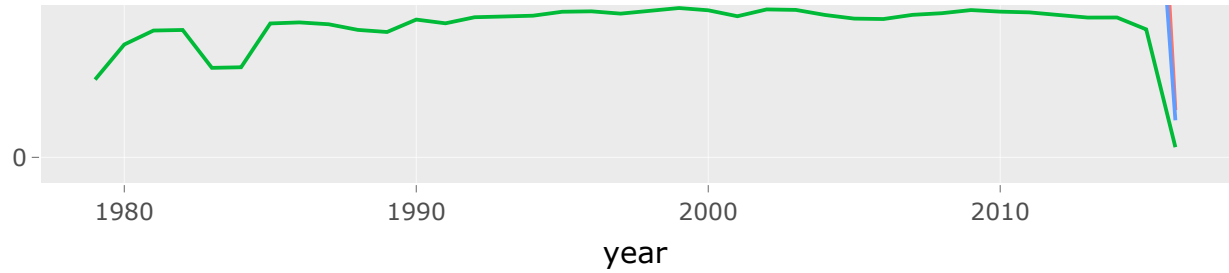
```
inner_join$value <- as.numeric(inner_join$value)
left_join$value <- as.numeric(left_join$value)

gender_data_2 <- left_join |>
    group_by(year, sex) |>
    summarise(
      total_suicides = sum(suicides_no, na.rm = TRUE),
      ave_suicide_rate = mean(value,na.rm = TRUE)
    )
```

```
`summarise()` has grouped output by 'year'. You can override using the
`.groups` argument.
```

```
plot1 <- ggplot(gender_data_2) +
    geom_line(mapping = aes(
    x = year,
    y = total_suicides,
    color = sex,
    ))
interactive_line_chart <- ggplotly(plot1, tooltip = c("x", "y", "color"))
interactive_line_chart
```





Graph 1 illustrates the overall suicide number over the years, categorized by sex: “Both sexes,” “Male,” and “Female.” From the data, it is evident that suicide rates among males are significantly higher compared to females, as represented by the blue line. The “Both sexes” category (red line) closely follows the male trend, indicating males constitute the majority of total suicides. Female suicides (green line) remain consistently lower throughout the observed period. The graph shows a general increase in suicides for both sexes until the late 1990s or early 2000s, followed by a gradual decline. A sharp drop is observed at the end of the timeline, which might indicate a data recording issue or a sudden improvement in preventive measures.

Graph 2

```r
# Calculate the average suicide rate by country
developed_countries <- c(
  "United States", "Canada", "United Kingdom", "Germany", "France",
  "Italy", "Japan", "Australia", "New Zealand", "Sweden", "Norway",
  "Denmark", "Finland", "Netherlands", "Belgium", "Switzerland",
  "Austria", "Ireland", "Luxembourg", "Spain", "Portugal", "Greece",
  "South Korea", "Singapore", "Iceland"
)

developed_df <- left_join |>
  filter(
    !is.na(suicide_rate) &
      suicide_rate > 0 &
      country %in% developed_countries
  )

developed_suicide_rate_df <- developed_df |>
  filter(!is.na(suicide_rate)) |>
  group_by(country) |>
  summarise(average_suicide_rate = mean(suicide_rate, na.rm = TRUE)) |>
  arrange(desc(average_suicide_rate))

top_5_country <- head(developed_suicide_rate_df, 5)
print(top_5_country)
```

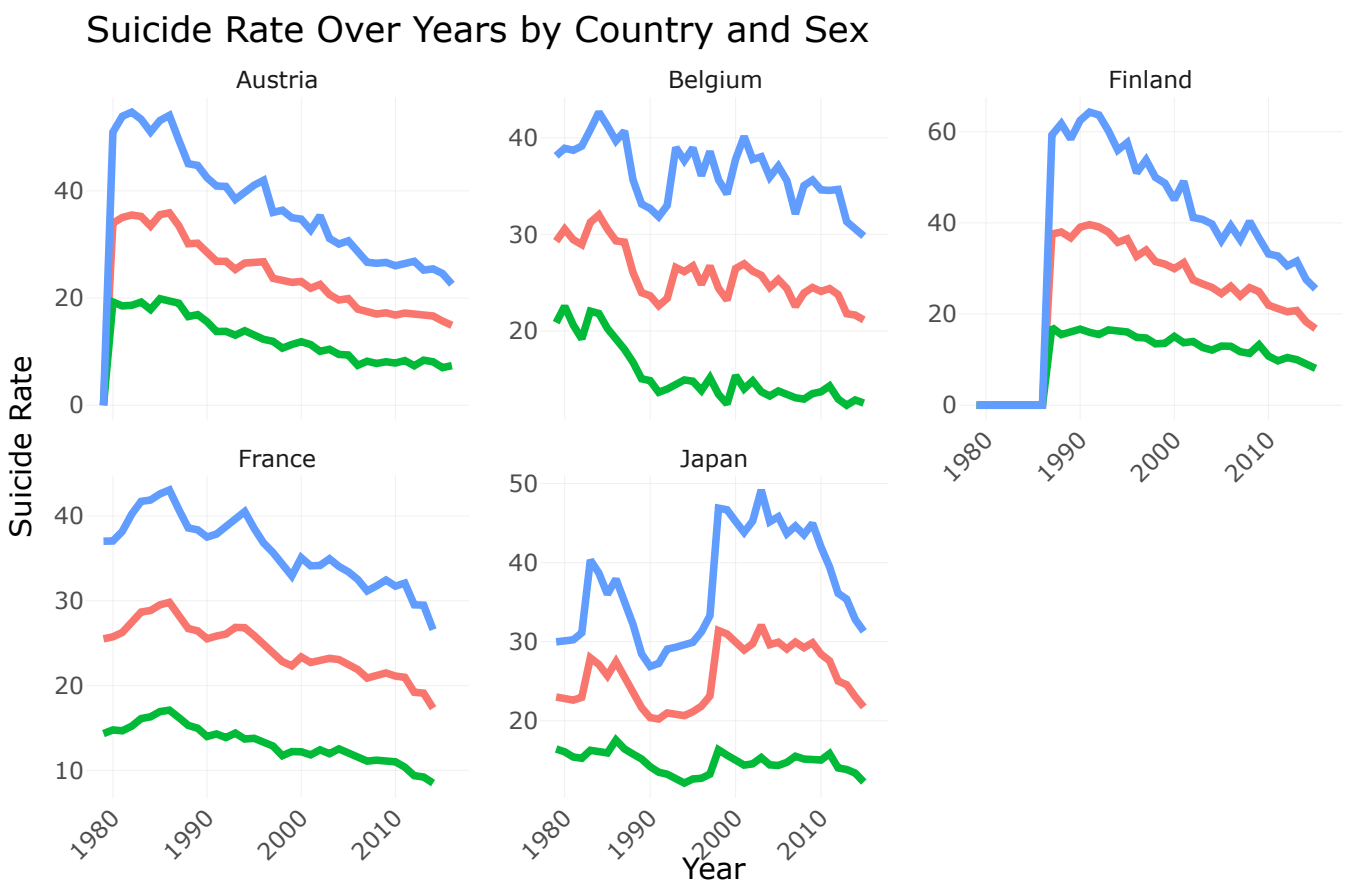
```
# A tibble: 5 × 2
  country average_suicide_rate
  <chr>                  <dbl>
1 Finland                 29.6
2 Belgium                 25.9
3 Japan                   25.8
4 Austria                 24.8
5 France                  24.5
```

```r
top_5_country_df <-
  left_join |>
  filter(country %in% c("Finland", "Belgium", "Japan", "Austria", "France")) %>%
  mutate(year = as.numeric(year))

plot2 <- ggplot(top_5_country_df, aes(x = year, y = suicide_rate, color = sex)) +
  geom_line(size = 1) +
  facet_wrap(~country, scales = "free_y") +
  labs(
    title = "Suicide Rate Over Years by Country and Sex",
    x = "Year",
    y = "Suicide Rate",
    color = "Sex"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.
```

```r
interactive_line_chart <- ggplotly(plot2, tooltip = c("x", "y", "color"))
interactive_line_chart
```



Graph 2 shows the suicide rates over time by country (Austria, Belgium, Finland, France, and Japan) and by sex (male, female, and both sexes). Across all countries, male suicide rates (blue lines) are

consistently higher than female rates (green lines), which remain relatively stable. Finland exhibits the highest male suicide rates in the early years, while Austria and Belgium show significant declines over time. Japan's male rates are notably volatile but remain consistently high. The "both sexes" category (red lines) follows male trends closely, underscoring the disproportionate contribution of male suicides to overall rates. This graph further highlights the sociocultural and economic factors influencing suicide rates across these nations. Japan's volatility may reflect societal pressures and economic shifts, while Finland's high rates suggest the impact of isolation and climate. The steady declines in Austria, Belgium, and France imply successful intervention strategies, though persistent gender gaps demand a focus on male-specific mental health policies.

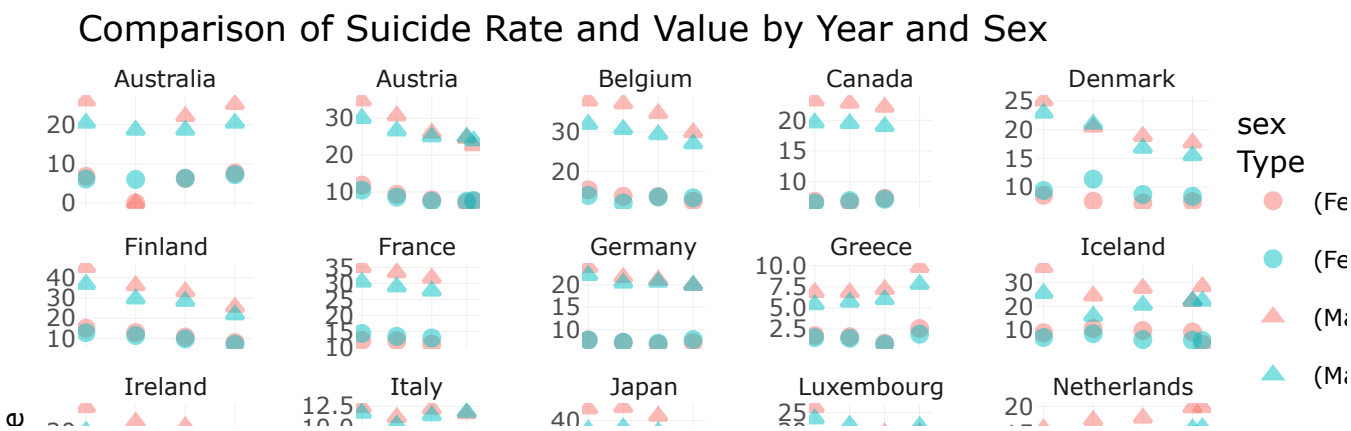
Graph 3

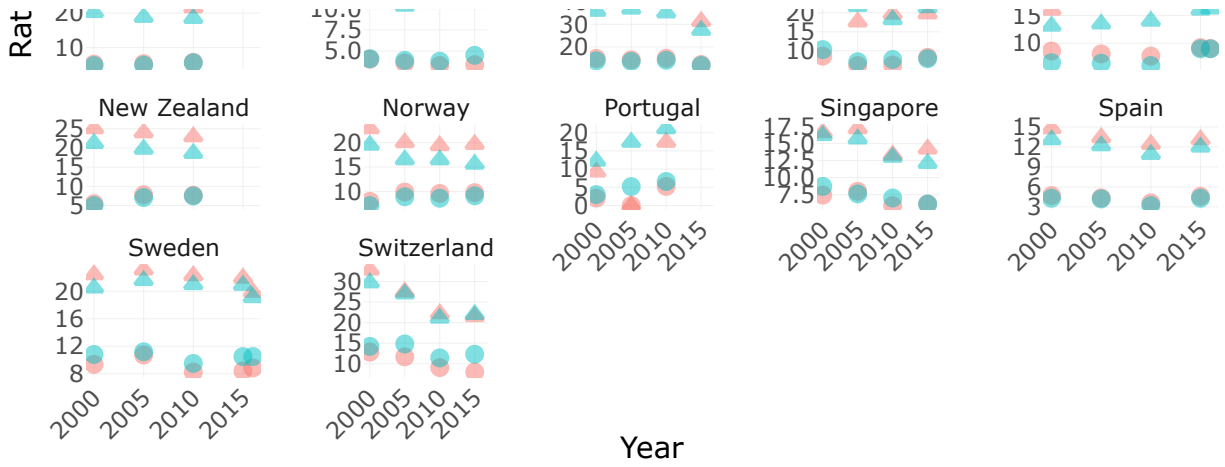
```
developed_inner_df <-
  inner_join |>
  filter(!is.na(suicide_rate) & !is.na(value),
         country %in% developed_countries
         )

long_developed_df <-
  developed_inner_df |>
  pivot_longer(cols = c(suicide_rate, value), names_to = "type", values_to = "rate")

plot3 <- ggplot(long_developed_df, aes(x = year, y = rate, size = sex, color = type))
  geom_point(aes(shape = sex), size = 2, alpha = 0.5) +
  facet_wrap(~country, scales = "free_y") +
  labs(
    title = "Comparison of Suicide Rate and Value by Year and Sex",
    x = "Year",
    y = "Rate",
    size = "Rate",
    color = "Type"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )

interactive_scatter_plot <- ggplotly(plot3, tooltip = c("x", "y", "size", "color"))
interactive_scatter_plot
```





Graph 3 compares suicide rates and values across various developed countries, categorized by year and sex. The data highlights significant gender disparities, with male suicide rates (represented by triangles) consistently higher than female rates (represented by circles) across all countries. Finland, Japan, and Iceland exhibit some of the highest male suicide rates, emphasizing the need for gender-targeted mental health policies. Countries like Greece and Portugal show lower rates overall, reflecting varying sociocultural or economic influences. The trend remains relatively stable over time for most nations, suggesting limited progress in reducing male-female disparities.

Discussion of Missing Data

The WHO Suicide Statistics dataset may have missing data due to underreporting, incomplete population counts, or gaps in country-level reporting for specific years or demographics, potentially skewing global trends. The Crude Suicide Rates dataset might have missing values from countries with inconsistent reporting or demographic subgroups, impacting temporal or cross-country comparisons. Addressing missing data could involve excluding incomplete records, aggregating data, or using interpolation to estimate gaps, ensuring analyses remain reliable and unbiased.

Limitations

The WHO Suicide Statistics dataset may have gaps due to underreporting or incomplete demographic data from certain countries and years. The Crude Suicide Rates dataset might miss values for specific demographics or years because of inconsistent reporting practices. These gaps could skew analyses, especially when comparing trends across countries or regions with differing data quality.

Conclusion

This project analyzes suicide rates globally, focusing on gender disparities, country-specific trends, and the factors influencing these rates. Sepecifically for developed country, by highlighting how developed nations like Finland, Belgium, Japan, Austria, and France are selected for their high average suicide rates, despite their strong healthcare systems and high standards of living. Male suicide rates are consistently higher than female rates, reflecting underlying gender-specific factors such as societal expectations, mental health stigma, and economic pressures that is always under researched. As for future data analyzing it is crucial to be aware of historical remains on data biases.

This project emphasizes the importance of understanding these trends in high-risk countries. For instance, Japan's high rates are linked to cultural norms prioritizing performance and conformity, while Finland's challenges include isolation and long winters, which can exacerbate mental health

issues. Belgium, Austria, and France show gradual declines, suggesting that intervention strategies like improved access to mental health resources and public awareness campaigns can be effective. Analyzing these patterns is crucial for developing gender-specific counseling and support systems to reduce suicide rates globally.

Additionally, despite differences in data collection methods across institutions and countries, the trends are strikingly similar, underscoring the global relevance of this issue. This consistency highlights the universal need for comprehensive mental health strategies that address gender disparities and societal influences. By focusing on the top five countries and their specific contexts, this assignment provides insights into the underlying causes and potential solutions to reduce suicide rates worldwide.